CARDLECT SYSTEM DOCUMENTATION

A FOLLOWING PAGE IS EMPTY – To make this document 15 pages.

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By

Cardlect **Buildlect**

Ooh! It’s THE NEXT PAGE EMPTY…

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CARDLECT SYSTEM DOCUMENTATION

by Buildlect Technologies Ltd.

Smart Cards. Smarter Schools

Version: 1.0 | Last Updated: 14-Nov-2025

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| 1. Introduction

Cardlect is a complete smart card ecosystem designed to simplify, secure, and centralise school operations. It connects students, parents, and staff to attendance, gate access, wallet transactions, library services, clinic records, event validation, live image verification, visitor handling, and school-wide analytics. Visitors receive temporary cards with strictly controlled permissions. The platform supports multi-school tenancy where each school operates on its own subdomain, ensuring full isolation, security, and independent control.

Parents have dedicated dashboards where they can deposit funds, view all histories, credit their children, and optionally deposit directly for a child through the Admin. Cardlect does not handle or store cash. All deposits are made to and processed by school finance staff. Cardlect records and manages wallet balances digitally.

Student cards are enabled for purchases inside school stores. Parent cards are used for pickup, and parents may optionally turn on a setting that allows their own parent card to be used for purchases. All features operate under a structured activation system. The Super Admin enables features for each school. The School Admin activates only the features permitted. Staff and parents may toggle selected non-critical options. All core operational functions remain school-controlled.

The system uses USB NFC readers as its primary scanning devices. Optional USB cameras support live image verification and QR or barcode scans. Fast Scan Mode allows instant operations across supported roles without needing full page interactions. Live image verification works for students, parents, and visitors during regular scans and in Fast Scan. Super Admin and School Admin have dedicated login systems separate from general staff and parents.

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| 2. System Architecture

| Frontend
Public pages such as landing, privacy, and terms use HTML, Tailwind CSS, and JavaScript. Dashboards for Admin, Staff, and Parents are built with React and Tailwind CSS. The system supports NFC scans, USB camera scans, mobile camera scans, and live image verification. Mobile scanning defaults to the back camera unless NFC is supported and explicitly enabled.

| Backend
The backend uses Node.js with Express, with all features available through a unified API at buildlect.com/api/cardlect.

| Database
PostgreSQL is used with tables for students, parents, staff, cards, attendance, gate logs, wallets, transactions, library records, clinic entries, event logs, visitor cards, hardware configurations, feature toggles, permissions, notifications, and audit logs. All data is isolated by school. Sensitive data is encrypted. Backups run automatically.

| Hardware
Supported hardware includes USB NFC readers, USB cameras, student cards, parent cards, and visitor cards. Hardware pricing uses fixed rates or the latest market prices when supplied.

| Diagrams & Flowcharts
Visual diagrams for system architecture, card workflows, and Fast Scan processes will be used for training and quick reference.

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| 3. User Roles and Access

| Super Admin
Separate login. Oversees all schools, features, hardware, analytics, configurations, and audit logs. Can view global reports across all schools.

| School Admin
Manages students, staff, parents, classes, cards, wallets, hardware preferences, and school settings. Activates only the features permitted by the Super Admin. Can configure role-based access and notifications per school.

| Staff Roles
Roles include Security, Teacher, Storekeeper, Librarian, Clinic Staff, Event Staff, and Accountant. Each role sees only the tools needed for its duties. Role permissions can be adjusted by the School Admin within the limits granted to the school. Optional live verification is supported per staff scan.

| Parent
Parents have full dashboards. They deposit funds through the Admin, credit children, view histories, freeze or unfreeze child wallets, transfer funds to other parents in the same school, and transfer across schools when allowed by both schools. Parents can also toggle whether their own parent card can be used for purchases. Live verification can be enabled.

| Student
Students do not log into any dashboard. They use their student cards for all school functions. Live verification ensures proper identification.

| Visitor
Visitors receive temporary cards with limited permissions. Entry and exit are logged and monitored. Live verification can be optionally enabled. Alerts can notify Admins of unusual visitor activity.

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| 4. Smart Card Logic

| Student Cards
Attendance, gate entry and exit, store purchases, library operations, clinic access, event validation, and any linked school service. Supports live image verification. Optional offline verification available in case of connectivity issues.

| Parent Cards
Pickup, optional live image or PIN verification, and optional purchases if the parent enables purchase mode. Notifications sent for purchases and fund activity.

| Visitor Cards
Controlled gate access only. Supports live image verification when enabled. Temporary card expiry enforced.

| Staff Scanning Logic
When a staff member scans a card, the action is processed based on their role, ensuring correct workflows for attendance, gate, store, library, clinic, event operations, and live verification. Fast Scan operations are supported where enabled.

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| 5. Core Features and Workflows

| Student Registration
Handled by Super Admin or School Admin. Includes linking parents, issuing cards, setting classes, and optional live image capture for verification.

| Parent Registration
Created automatically during student registration or added manually. Card issued and wallet activated. Live image verification can be enabled.

| Parent Wallet Funding
Parents deposit through the Admin. Funds enter the parent wallet. From there, parents can credit their children or deposit directly to a child at the Admin office if permitted. All transactions are fully logged. Optional notifications alert parents to deposits and wallet activity.

| Wallet Spending
Students can spend at school stores unless their wallet is frozen. Parents can spend only when they activate the purchase toggle for their own card. Spending may include optional limits and notifications.

| Attendance
Teachers scan student cards. The system compares timestamps with scheduled class times and automatically marks lateness. Optional reasons may be recorded. Parents view attendance on their dashboards. Live verification ensures authenticity.

| Gate and Pickup
Security staff scan cards to log entry and exit. Parent pickup uses parent cards with optional PIN or live image verification based on school settings. Fast Scan supports rapid gate operations with instant verification. Alerts can notify Admins of irregular entries.

| Library
Book issuing and returns are handled by scanning cards. Due dates, fines, clearance processes, and live verification are automated.

| Clinic
Clinic staff scan cards to record visits and notes. Parents view clinic history. Live verification ensures proper identification. Notifications can alert parents to clinic events.

| Events
Event entry is validated through card scans. Event restrictions are configurable. Live verification optional. Admins can monitor attendance logs.

| Visitor Management
Visitors receive temporary cards. Entry and exit logs are captured automatically. Live verification can be enabled. Notifications sent to Admins for unusual activity.

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| 6. Dashboards

| Parent Dashboard
Parents can:
• View wallet balance
• Deposit through Admin
• Credit children
• Transfer to other parents
• Freeze or unfreeze child wallets
• Toggle parent card purchase
• Enable or disable live verification
• View full transaction and activity history
• Receive notifications and alerts

| Admin Dashboard
Admins manage student data, staff, parents, class lists, card management, attendance, gate logs, library, clinic, events, visitor logs, wallet finances, and school-level settings. Supports enabling live verification and Fast Scan controls. Alerts and notifications can be configured here. Reports exportable in CSV or PDF.

| Staff Dashboard
Each staff role receives an interface based on duties. Security sees gate functions. Teachers see attendance. Storekeepers see purchases. Librarians see book management. Clinic staff see health records. Event staff handle event entry. Live verification optional per role.

| Visual Aids
Screenshots or sample dashboard layouts recommended for staff training and onboarding.

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| 7. Fast Scan Feature

Fast Scan Mode performs instant processing for supported operations without loading forms or pages. Works on desktop and mobile, using NFC or camera. Supports live image verification and optional notifications. Can be enabled or disabled per role and device.

| Fast Scan Workflow
Fast Scan enabled
Card scanned
Action processed immediately
Optional live image captured
Record created instantly

Supported for attendance, gate entry and exit, store purchases, library operations, clinic visits, events, visitor handling, and verification workflows. Offline Fast Scan supported when connectivity is limited.

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| 8. Feature Activation and Toggle System

Super Admin controls high-level features for each school.
School Admin activates features available to the school.
Parents and staff may toggle non-critical features allowed by the school.

| Toggleable Features Include
• Parent card purchase mode
• Child wallet freeze and unfreeze
• Parent to parent transfer
• Parent cross-school transfer
• Fast Scan
• USB camera scanning
• Mobile NFC scanning
• Live image verification
• Library penalties
• Store spending limits
• Clinic visibility rules
• Visitor card activation
• Pickup PIN requirement
• Pickup live verification requirement
• Device-specific scanner selection
• Gate notifications and alerts
• Activity visibility settings
• Role-based quick scan permissions
• School-level controls on card actions
• Optional guardian approval rules
• Auto alerts for unusual activity
• Custom notifications for wallet or gate events
• Session-based scanning preferences per device
• Offline verification toggle
• Minimum hardware requirement toggle

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| 9. Multi-School Tenancy

Each school operates on a separate subdomain.
All data is isolated and cannot be accessed by another school.
Super Admin has a unified global dashboard for multi-school oversight. Analytics and logs can aggregate multiple schools for Super Admin view only.

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| 10. Hardware Integration

USB NFC readers are required for desktop scanning.
USB cameras are optional but recommended for live verification and scanning.
Mobile scanning works through camera or NFC if supported.
Hardware costs follow fixed prices or the latest market rates. Minimum camera specifications and lighting requirements recommended for accurate live verification. Optional secondary hardware can be added per school or device.

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| 11. Security and Compliance

• Role-based access control
• Encrypted storage
• Complete audit logs
• Daily backups
• Row-level multi-school isolation
• Optional live verification and PIN
• IP logging for sensitive actions
• Session timeout and auto logout
• Device registration and monitoring
• Card usage restrictions by location
• Controlled feature activation
• Secure QR and barcode scanning
• Alert logging for failed verifications
• Compliance with local data protection regulations

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| 12. API and Integration

API Base URL: https://buildlect.com/cardlect/api
Secured with API keys.
Developers can access full API documentation at https://buildlect.com/cardlect/devs.
Supports integrations for schools and approved third parties.
To confirm originality of this or any older version, refer to https://buildlect.com/cardlect/old.

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| 13. Logs, Reports, and Analytics

Logs include attendance, gate activity, wallet transactions, library usage, clinic visits, event attendance, visitor activity, device usage, and staff actions. Reports may be exported in PDF or CSV. Analytics provide insights into behaviour trends, financial patterns, attendance performance, verification events, and operational efficiency. Custom alert and notification logs included.

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| 14. Cardlect Fee Plan for Partner Schools

| Setup and Activation
Staff training: ₦75,000
Installation: ₦500 per student
Example (100 students): ₦125,000

| Hardware (fixed rate or latest market price)
USB NFC Reader (2 units): ₦70,000 each
USB Camera: ₦25,000
Student Cards (10+): ₦2,000 each
Parent Cards (2): ₦2,000 each
Visitor Cards (pack of 10): ₦20,000
Minimum hardware total: ₦209,000 or latest market values

| Monthly Maintenance Fee per Student
1 to 500 students: ₦200
501 to 1500 students: ₦150
1501 and above: ₦100

| Annual equivalents: ₦2,400, ₦1,800, ₦1,200
Example (100 students): ₦20,000 monthly, ₦240,000 yearly

| Optional Add-Ons
Extra NFC Reader: ₦70,000
Extra USB Camera: ₦25,000
Extra Smart Cards: ₦2,000 each
Branded Card Layout: ₦2,500 each
On-site Visit: ₦10,000 plus transport

| Warranty and Support
One month hardware warranty
Free remote support and updates
Paid on-site maintenance available

| Sample First-Year Estimate (100 Students)
Setup: ₦125,000
Hardware: ₦209,000
Annual Maintenance: ₦240,000
Total: ₦574,000
Cost per student per year excluding hardware: ₦2,900

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| 15. Future Development

Mobile apps
Offline support
Biometrics
Transport management
Hostel management
Unified billing
Computer-based testing
Behaviour and discipline tracking
Live verification AI enhancements
Integration with IoT-enabled devices
Advanced notifications and alerts
Offline verification options
Secondary hardware support

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| 16. Glossary

Roles: Super Admin, School Admin, Staff, Parent, Student, Visitor
Cards: Student Card, Parent Pickup Card, Visitor Card
Wallet Flow: Parent deposit to parent wallet then to child wallet
Intra-Transfer: Parent to parent within same school
Inter-Transfer: Parent to parent across schools
Inter-Transfer Control: Admin selects permitted schools
Feature Activation: Super Admin enables, Admin activates, others toggle permitted options
Sub-domains: Each school receives a unique system space
Live Verification: Real-time identity confirmation during scanning
Notifications: Alerts and logs generated for wallet, gate, or verification events
Minimum Hardware: Recommended camera resolution and lighting for live verification
Offline Verification: Enabled when connectivity is limited

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END OF CARDLECT SYSTEM DOCUMENTATION

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